

Series

If we try to add the terms of an infinite sequence $\{a_n\}_{n=1}^{\infty}$ we get an expression of the form

$$\boxed{1} \quad a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

which is called an **infinite series** (or just a **series**) and is denoted, for short, by the symbol

$$\sum_{n=1}^{\infty} a_n \quad \text{or} \quad \sum a_n$$

But does it make sense to talk about the sum of infinitely many terms?

It would be impossible to find a finite sum for the series

$$1 + 2 + 3 + 4 + 5 + \cdots + n + \cdots$$

because if we start adding the terms we get the cumulative sums 1, 3, 6, 10, 15, 21, ... and, after the n th term, we get $n(n+1)/2$, which becomes very large as n increases.

However, if we start to add the terms of the series

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \cdots + \frac{1}{2^n} + \cdots$$

we get $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \frac{31}{32}, \frac{63}{64}, \dots, 1 - 1/2^n, \dots$. The table shows that as we add more and more terms, these *partial sums* become closer and closer to 1.

In fact, by adding sufficiently many terms of the series we can make the partial sums as close as we like to 1. So it seems reasonable to say that the sum of this infinite series is 1 and to write

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots + \frac{1}{2^n} + \cdots = 1$$

We use a similar idea to determine whether or not a general series (1) has a sum. We consider the **partial sums**

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$s_4 = a_1 + a_2 + a_3 + a_4$$

and, in general,

$$s_n = a_1 + a_2 + a_3 + \cdots + a_n = \sum_{i=1}^n a_i$$

n	Sum of first n terms
1	0.50000000
2	0.75000000
3	0.87500000
4	0.93750000
5	0.96875000
6	0.98437500
7	0.99218750
10	0.99902344
15	0.99996948
20	0.99999905
25	0.99999997

These partial sums form a new sequence $\{s_n\}$, which may or may not have a limit. If $\lim_{n \rightarrow \infty} s_n = s$ exists (as a finite number), then, as in the preceding example, we call it the sum of the infinite series $\sum a_n$.

2 Definition Given a series $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$, let s_n denote its n th partial sum:

$$s_n = \sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n$$

If the sequence $\{s_n\}$ is convergent and $\lim_{n \rightarrow \infty} s_n = s$ exists as a real number, then the series $\sum a_n$ is called **convergent** and we write

$$a_1 + a_2 + \cdots + a_n + \cdots = s \quad \text{or} \quad \sum_{n=1}^{\infty} a_n = s$$

The number s is called the **sum** of the series. Otherwise, the series is called **divergent**.

Thus, when we write $\sum_{n=1}^{\infty} a_n = s$ we mean that by adding sufficiently many terms of the series we can get as close as we like to the number s . Notice that

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i$$

EXAMPLE 1 An important example of an infinite series is the geometric series

$$a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1} \quad a \neq 0$$

Each term is obtained from the preceding one by multiplying it by the common ratio r .

If $r = 1$, then $s_n = a + a + \cdots + a = na \rightarrow \pm\infty$. Since $\lim_{n \rightarrow \infty} s_n$ doesn't exist, the geometric series diverges in this case.

If $r \neq 1$, we have

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1}$$

and

$$rs_n = ar + ar^2 + \cdots + ar^{n-1} + ar^n$$

Subtracting these equations, we get

$$s_n - rs_n = a - ar^n$$

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$$s_n = \frac{a(1 - r^n)}{1 - r}$$

If $-1 < r < 1$, we know from [Equation 2](#) that $r^n \rightarrow 0$ as $n \rightarrow \infty$, so

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \frac{a(1 - r^n)}{1 - r} = \frac{a}{1 - r} - \frac{a}{1 - r} \lim_{n \rightarrow \infty} r^n = \frac{a}{1 - r}$$

Thus, when $|r| < 1$ the geometric series is convergent and its sum is $a/(1 - r)$.

If $r \leq -1$ or $r > 1$, the sequence $\{r^n\}$ is divergent [\(see Problem 19\)](#) and so, by Equation 3, $\lim_{n \rightarrow \infty} s_n$ does not exist. Therefore, the geometric series diverges in those cases.

4 The geometric series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \dots$$

is convergent if $|r| < 1$ and its sum is

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} \quad |r| < 1$$

If $|r| \geq 1$, the geometric series is divergent.

 In words: The sum of a convergent geometric series is

$$\frac{\text{first term}}{1 - \text{common ratio}}$$

EXAMPLE 2 Find the sum of the geometric series

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \cdots$$

SOLUTION The first term is $a = 5$ and the common ratio is $r = -\frac{2}{3}$. Since $|r| = \frac{2}{3} < 1$, the series is convergent by (4) and its sum is

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \cdots = \frac{5}{1 - (-\frac{2}{3})} = \frac{5}{\frac{5}{3}} = 3$$

EXAMPLE 3 Is the series $\sum_{n=1}^{\infty} 2^{2n}3^{1-n}$ convergent or divergent?

SOLUTION Let's rewrite the n th term of the series in the form ar^{n-1} :

$$\sum_{n=1}^{\infty} 2^{2n}3^{1-n} = \sum_{n=1}^{\infty} \frac{4^n}{3^{n-1}} = \sum_{n=1}^{\infty} 4\left(\frac{4}{3}\right)^{n-1}$$

We recognize this series as a geometric series with $a = 4$ and $r = \frac{4}{3}$. Since $r > 1$, the series diverges by (4).

The geometric series with $a = 1/9$ and $r = 1/3$ is

$$\frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \dots = \sum_{n=1}^{\infty} \frac{1}{9} \left(\frac{1}{3}\right)^{n-1} = \frac{1/9}{1 - (1/3)} = \frac{1}{6}.$$

The series

$$\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n} = 5 - \frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots$$

is a geometric series with $a = 5$ and $r = -1/4$. It converges to

$$\frac{a}{1 - r} = \frac{5}{1 + (1/4)} = 4.$$

EXAMPLE 4 Write the number $2.\overline{317} = 2.3171717\ldots$ as a ratio of integers.

SOLUTION

$$2.3171717\ldots = 2.3 + \frac{17}{10^3} + \frac{17}{10^5} + \frac{17}{10^7} + \cdots$$

After the first term we have a geometric series with $a = 17/10^3$ and $r = 1/10^2$.
Therefore

$$\begin{aligned} 2.\overline{317} &= 2.3 + \frac{\frac{17}{10^3}}{1 - \frac{1}{10^2}} = 2.3 + \frac{\frac{17}{1000}}{\frac{99}{100}} \\ &= \frac{23}{10} + \frac{17}{990} = \frac{1147}{495} \end{aligned}$$

Express the repeating decimal $5.232323 \dots$ as the ratio of two integers.

Solution From the definition of a decimal number, we get a geometric series

$$\begin{aligned} 5.232323 \dots &= 5 + \frac{23}{100} + \frac{23}{(100)^2} + \frac{23}{(100)^3} + \dots \\ &= 5 + \frac{23}{100} \underbrace{\left(1 + \frac{1}{100} + \left(\frac{1}{100} \right)^2 + \dots \right)}_{1/(1 - 0.01)} \quad \begin{array}{l} a = 1, \\ r = 1/100 \end{array} \\ &= 5 + \frac{23}{100} \left(\frac{1}{0.99} \right) = 5 + \frac{23}{99} = \frac{518}{99} \end{aligned}$$

(Examples of geometric series and their sums)

$$(a) \quad 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} = \frac{1}{1 - \frac{1}{2}} = 2. \quad \text{Here } a = 1 \text{ and } r = \frac{1}{2}.$$

Since $|r| < 1$, the series converges.

$$(b) \quad \pi - e + \frac{e^2}{\pi} - \frac{e^3}{\pi^2} + \cdots = \sum_{n=1}^{\infty} \pi \left(-\frac{e}{\pi}\right)^{n-1} \quad \text{Here } a = \pi \text{ and } r = -\frac{e}{\pi}.$$
$$= \frac{\pi}{1 - \left(-\frac{e}{\pi}\right)} = \frac{\pi^2}{\pi + e}.$$

The series converges since $\left|-\frac{e}{\pi}\right| < 1$.

$$(c) \quad 1 + 2^{1/2} + 2 + 2^{3/2} + \cdots = \sum_{n=1}^{\infty} (\sqrt{2})^{n-1}. \quad \text{This series diverges to } \infty \text{ since } a = 1 > 0$$

and $r = \sqrt{2} > 1$.

$$(d) \quad 1 - 1 + 1 - 1 + 1 - \cdots = \sum_{n=1}^{\infty} (-1)^{n-1}. \quad \text{This series diverges since } r = -1.$$

(e) Let $x = 0.32\,32\,32\,\cdots = 0.\overline{32}$; then

$$x = \frac{32}{100} + \frac{32}{100^2} + \frac{32}{100^3} + \cdots = \sum_{n=1}^{\infty} \frac{32}{100} \left(\frac{1}{100}\right)^{n-1} = \frac{32}{100} \frac{1}{1 - \frac{1}{100}} = \frac{32}{99}.$$

EXAMPLE 5 Find the sum of the series $\sum_{n=0}^{\infty} x^n$, where $|x| < 1$.

SOLUTION Notice that this series starts with $n = 0$ and so the first term is $x^0 = 1$. (With series, we adopt the convention that $x^0 = 1$ even when $x = 0$.) Thus

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \cdots$$

This is a geometric series with $a = 1$ and $r = x$. Since $|r| = |x| < 1$, it converges and (4) gives

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$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}$$

EXAMPLE 6 Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ is convergent, and find its sum.

SOLUTION This is not a geometric series, so we go back to the definition of a convergent series and compute the partial sums.

$$s_n = \sum_{i=1}^n \frac{1}{i(i+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)}$$

We can simplify this expression if we use the partial fraction decomposition

$$\frac{1}{i(i+1)} = \frac{1}{i} - \frac{1}{i+1}$$

Thus, we have

$$\begin{aligned} s_n &= \sum_{i=1}^n \frac{1}{i(i+1)} = \sum_{i=1}^n \left(\frac{1}{i} - \frac{1}{i+1} \right) \\ &= \left(1 - \cancel{\frac{1}{2}} \right) + \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} \right) + \left(\cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} \right) + \cdots + \left(\cancel{\frac{1}{n}} - \frac{1}{n+1} \right) \\ &= 1 - \frac{1}{n+1} \end{aligned}$$

and so

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) = 1 - 0 = 1$$

Therefore, the given series is convergent and

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

EXAMPLE 7 Show that the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

is divergent.

SOLUTION

$$s_1 = 1$$

$$s_2 = 1 + \frac{1}{2}$$

$$s_4 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = 1 + \frac{2}{2}$$

$$s_8 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)$$

$$> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)$$

$$= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 1 + \frac{3}{2}$$

$$s_{16} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right)$$

$$> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \cdots + \frac{1}{8}\right) + \left(\frac{1}{16} + \cdots + \frac{1}{16}\right)$$

$$= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 1 + \frac{4}{2}$$

Similarly, $s_{32} > 1 + \frac{5}{2}$, $s_{64} > 1 + \frac{6}{2}$, and in general

$$s_{2^n} > 1 + \frac{n}{2}$$

This shows that $s_{2^n} \rightarrow \infty$ as $n \rightarrow \infty$ and so $\{s_n\}$ is divergent. Therefore, the harmonic series diverges.

6 Theorem If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof Let $s_n = a_1 + a_2 + \cdots + a_n$. Then $a_n = s_n - s_{n-1}$. Since $\sum a_n$ is convergent, the sequence $\{s_n\}$ is convergent. Let $\lim_{n \rightarrow \infty} s_n = s$. Since $n - 1 \rightarrow \infty$ as $n \rightarrow \infty$, we also have $\lim_{n \rightarrow \infty} s_{n-1} = s$. Therefore

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} (s_n - s_{n-1}) = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} \\ &= s - s = 0\end{aligned}$$

NOTE 1 • With any *series* $\sum a_n$ we associate two *sequences*: the sequence $\{s_n\}$ of its partial sums and the sequence $\{a_n\}$ of its terms. If $\sum a_n$ is convergent, then the limit of the sequence $\{s_n\}$ is s (the sum of the series) and, as Theorem 6 asserts, the limit of the sequence $\{a_n\}$ is 0.

NOTE 2 • The converse of Theorem 6 is not true in general. If $\lim_{n \rightarrow \infty} a_n = 0$, we cannot conclude that $\sum a_n$ is convergent. Observe that for the harmonic series $\sum 1/n$ we have $a_n = 1/n \rightarrow 0$ as $n \rightarrow \infty$, but we showed in Example 7 that $\sum 1/n$ is divergent.

7 The Test for Divergence If $\lim_{n \rightarrow \infty} a_n$ does not exist or if $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

The Test for Divergence follows from Theorem 6 because, if the series is not divergent, then it is convergent, and so $\lim_{n \rightarrow \infty} a_n = 0$.

(a) $\sum_{n=1}^{\infty} n^2$ diverges because $n^2 \rightarrow \infty$.

(b) $\sum_{n=1}^{\infty} \frac{n+1}{n}$ diverges because $\frac{n+1}{n} \rightarrow 1$. $\lim_{n \rightarrow \infty} a_n \neq 0$

(c) $\sum_{n=1}^{\infty} (-1)^{n+1}$ diverges because $\lim_{n \rightarrow \infty} (-1)^{n+1}$ does not exist.

(d) $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$ diverges because $\lim_{n \rightarrow \infty} \frac{-n}{2n+5} = -\frac{1}{2} \neq 0$.

(a) $\sum_{n=1}^{\infty} \frac{n}{2n-1}$ diverges to infinity since $\lim_{n \rightarrow \infty} \frac{n}{2n-1} = 1/2 > 0$.

(b) $\sum_{n=1}^{\infty} (-1)^n n \sin(1/n)$ diverges since

$$\lim_{n \rightarrow \infty} \left| (-1)^n n \sin \frac{1}{n} \right| = \lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n} = \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1 \neq 0.$$

EXAMPLE 8 Show that the series $\sum_{n=1}^{\infty} \frac{n^2}{5n^2 + 4}$ diverges.

SOLUTION

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n^2}{5n^2 + 4} = \lim_{n \rightarrow \infty} \frac{1}{5 + 4/n^2} = \frac{1}{5} \neq 0$$

So the series diverges by the Test for Divergence.

NOTE 3 If we find that $\lim_{n \rightarrow \infty} a_n \neq 0$, we know that $\sum a_n$ is divergent. If we find that $\lim_{n \rightarrow \infty} a_n = 0$, we know *nothing* about the convergence or divergence of $\sum a_n$. Remember the warning in Note 2: If $\lim_{n \rightarrow \infty} a_n = 0$, the series $\sum a_n$ might converge or it might diverge.

8 Theorem If $\sum a_n$ and $\sum b_n$ are convergent series, then so are the series $\sum ca_n$ (where c is a constant), $\sum (a_n + b_n)$, and $\sum (a_n - b_n)$, and

$$\begin{aligned} \text{(i)} \quad \sum_{n=1}^{\infty} ca_n &= c \sum_{n=1}^{\infty} a_n & \text{(ii)} \quad \sum_{n=1}^{\infty} (a_n + b_n) &= \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n \\ \text{(iii)} \quad \sum_{n=1}^{\infty} (a_n - b_n) &= \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n \end{aligned}$$

EXAMPLE 9 Find the sum of the series $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right)$.

SOLUTION The series $\sum 1/2^n$ is a geometric series with $a = \frac{1}{2}$ and $r = \frac{1}{2}$, so

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1$$

In Example 6 we found that

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

So, by Theorem 8, the given series is convergent and

$$\begin{aligned} \sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right) &= 3 \sum_{n=1}^{\infty} \frac{1}{n(n+1)} + \sum_{n=1}^{\infty} \frac{1}{2^n} \\ &= 3 \cdot 1 + 1 = 4 \end{aligned}$$

NOTE 4 • A finite number of terms doesn't affect the convergence or divergence of a series. For instance, suppose that we were able to show that the series

$$\sum_{n=4}^{\infty} \frac{n}{n^3 + 1}$$

is convergent. Since

$$\sum_{n=1}^{\infty} \frac{n}{n^3 + 1} = \frac{1}{2} + \frac{2}{9} + \frac{3}{28} + \sum_{n=4}^{\infty} \frac{n}{n^3 + 1}$$

it follows that the entire series $\sum_{n=1}^{\infty} n/(n^3 + 1)$ is convergent. Similarly, if it is known that the series $\sum_{n=N+1}^{\infty} a_n$ converges, then the full series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^N a_n + \sum_{n=N+1}^{\infty} a_n$$

is also convergent.

EXAMPLE 8 The series

$$1 + \underbrace{\frac{1}{2} + \frac{1}{2}}_{2 \text{ terms}} + \underbrace{\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}}_{4 \text{ terms}} + \cdots + \underbrace{\frac{1}{2^n} + \frac{1}{2^n} + \cdots + \frac{1}{2^n}}_{2^n \text{ terms}} + \cdots$$

diverges because the terms can be grouped into infinitely many clusters each of which adds to 1, so the partial sums increase without bound. However, the terms of the series form a sequence that converges to 0.]

1. Every nonzero constant multiple of a divergent series diverges.
2. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverge.

Caution Remember that $\sum(a_n + b_n)$ can converge when $\sum a_n$ and $\sum b_n$ both diverge. For example, $\sum a_n = 1 + 1 + 1 + \cdots$ and $\sum b_n = (-1) + (-1) + (-1) + \cdots$ diverge, whereas $\sum(a_n + b_n) = 0 + 0 + 0 + \cdots$ converges to 0.

Find the sums of the following series.

$$\begin{aligned} \text{(a)} \quad \sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}} &= \sum_{n=1}^{\infty} \left(\frac{1}{2^{n-1}} - \frac{1}{6^{n-1}} \right) \\ &= \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} - \sum_{n=1}^{\infty} \frac{1}{6^{n-1}} && \text{Difference Rule} \\ &= \frac{1}{1 - (1/2)} - \frac{1}{1 - (1/6)} && \text{Geometric series with } a = 1 \text{ and } r = 1/2, 1/6 \\ &= 2 - \frac{6}{5} = \frac{4}{5} \end{aligned}$$

$$(b) \sum_{n=0}^{\infty} \frac{4}{2^n} = 4 \sum_{n=0}^{\infty} \frac{1}{2^n}$$

Constant Multiple Rule

$$= 4 \left(\frac{1}{1 - (1/2)} \right)$$

Geometric series with $a = 1, r = 1/2$

$$= 8$$

Find the sum of the series $\sum_{n=1}^{\infty} \frac{1 + 2^{n+1}}{3^n}$.

Solution The given series is the sum of two geometric series,

$$\sum_{n=1}^{\infty} \frac{1}{3^n} = \sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{1}{3}\right)^{n-1} = \frac{1/3}{1 - (1/3)} = \frac{1}{2} \quad \text{and}$$

$$\sum_{n=1}^{\infty} \frac{2^{n+1}}{3^n} = \sum_{n=1}^{\infty} \frac{4}{3} \left(\frac{2}{3}\right)^{n-1} = \frac{4/3}{1 - (2/3)} = 4.$$

Thus, its sum is $\frac{1}{2} + 4 = \frac{9}{2}$

Adding or Deleting Terms

We can add a finite number of terms to a series or delete a finite number of terms without altering the series' convergence or divergence, although in the case of convergence this will usually change the sum. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$ and

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \cdots + a_{k-1} + \sum_{n=k}^{\infty} a_n.$$

Conversely, if $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$, then $\sum_{n=1}^{\infty} a_n$ converges. Thus,

$$\sum_{n=1}^{\infty} \frac{1}{5^n} = \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \sum_{n=4}^{\infty} \frac{1}{5^n}$$

and

$$\sum_{n=4}^{\infty} \frac{1}{5^n} = \left(\sum_{n=1}^{\infty} \frac{1}{5^n} \right) - \frac{1}{5} - \frac{1}{25} - \frac{1}{125}.$$

Reindexing

As long as we preserve the order of its terms, we can reindex any series without altering its convergence. To raise the starting value of the index h units, replace the n in the formula for a_n by $n - h$:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1+h}^{\infty} a_{n-h} = a_1 + a_2 + a_3 + \cdots.$$

To lower the starting value of the index h units, replace the n in the formula for a_n by $n + h$:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1-h}^{\infty} a_{n+h} = a_1 + a_2 + a_3 + \cdots.$$

We saw this reindexing in starting a geometric series with the index $n = 0$ instead of the index $n = 1$, but we can use any other starting index value as well. We usually give preference to indexings that lead to simple expressions.

We can write the geometric series

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{4} + \cdots$$

as

$$\sum_{n=0}^{\infty} \frac{1}{2^n}, \quad \sum_{n=5}^{\infty} \frac{1}{2^{n-5}}, \quad \text{or even} \quad \sum_{n=-4}^{\infty} \frac{1}{2^{n+4}}.$$

The partial sums remain the same no matter what indexing we choose.

The Integral Test

The Integral Test Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x) dx$ is convergent. In other words:

- (i) If $\int_1^{\infty} f(x) dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.
- (ii) If $\int_1^{\infty} f(x) dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

NOTE ^o When we use the Integral Test it is not necessary to start the series or the integral at $n = 1$. For instance, in testing the series

$$\sum_{n=4}^{\infty} \frac{1}{(n-3)^2} \quad \text{we use} \quad \int_4^{\infty} \frac{1}{(x-3)^2} dx$$

Also, it is not necessary that f be always decreasing. What is important is that f be *ultimately* decreasing, that is, decreasing for x larger than some number N . Then $\sum_{n=N}^{\infty} a_n$ is convergent, so $\sum_{n=1}^{\infty} a_n$ is convergent

EXAMPLE 1 Test the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$ for convergence or divergence.

SOLUTION The function $f(x) = 1/(x^2 + 1)$ is continuous, positive, and decreasing on $[1, \infty)$ so we use the Integral Test:

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^2 + 1} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2 + 1} dx = \lim_{t \rightarrow \infty} \tan^{-1} x \Big|_1^t \\ &= \lim_{t \rightarrow \infty} \left(\tan^{-1} t - \frac{\pi}{4} \right) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \end{aligned}$$

Thus, $\int_1^{\infty} 1/(x^2 + 1) dx$ is a convergent integral and so, by the Integral Test, the series $\sum 1/(n^2 + 1)$ is convergent.

EXAMPLE 2 For what values of p is the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ convergent?

SOLUTION If $p < 0$, then $\lim_{n \rightarrow \infty} (1/n^p) = \infty$. If $p = 0$, then $\lim_{n \rightarrow \infty} (1/n^p) = 1$. In either case $\lim_{n \rightarrow \infty} (1/n^p) \neq 0$, so the given series diverges by the Test for Divergence.

If $p > 0$, then the function $f(x) = 1/x^p$ is clearly continuous, positive, and decreasing on $[1, \infty)$. We found that

$$\int_1^{\infty} \frac{1}{x^p} dx \text{ converges if } p > 1 \text{ and diverges if } p \leq 1$$

It follows from the Integral Test that the series $\sum 1/n^p$ converges if $p > 1$ and diverges if $0 < p \leq 1$. (For $p = 1$, this series is the harmonic series.)

The series in Example 2 is called the p -series.

1 The p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leq 1$.

EXAMPLE 3

(a) The series

$$\sum_{n=1}^{\infty} \frac{1}{n^3} = \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \cdots$$

is convergent because it is a p -series with $p = 3 > 1$.

(b) The series

$$\sum_{n=1}^{\infty} \frac{1}{n^{1/3}} = \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}} = 1 + \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{3}} + \frac{1}{\sqrt[3]{4}} + \cdots$$

is divergent because it is a p -series with $p = \frac{1}{3} < 1$.

NOTE • We should *not* infer from the Integral Test that the sum of the series is equal to the value of the integral. In fact,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad \text{whereas} \quad \int_1^{\infty} \frac{1}{x^2} dx = 1$$

Therefore, in general,

$$\sum_{n=1}^{\infty} a_n \neq \int_1^{\infty} f(x) dx$$

EXAMPLE 4 Determine whether the series $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ converges or diverges.

SOLUTION The function $f(x) = (\ln x)/x$ is positive and continuous for $x > 1$ because the logarithm function is continuous. But it is not obvious whether or not f is decreasing, so we compute its derivative:

$$f'(x) = \frac{(1/x)x - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

Thus, $f'(x) < 0$ when $\ln x > 1$, that is, $x > e$. It follows that f is decreasing when $x > e$ and so we can apply the Integral Test:

$$\begin{aligned} \int_1^{\infty} \frac{\ln x}{x} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{\ln x}{x} dx = \lim_{t \rightarrow \infty} \left[\frac{(\ln x)^2}{2} \right]_1^t \\ &= \lim_{t \rightarrow \infty} \frac{(\ln t)^2}{2} = \infty \end{aligned}$$

Since this improper integral is divergent, the series $\sum (\ln n)/n$ is also divergent by the Integral Test.

The Comparison Tests

The Comparison Test Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

- (i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n , then $\sum a_n$ is also convergent.
- (ii) If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n , then $\sum a_n$ is also divergent.

EXAMPLE 1 Determine whether the series $\sum_{n=1}^{\infty} \frac{5}{2n^2 + 4n + 3}$ converges or diverges.

SOLUTION For large n the dominant term in the denominator is $2n^2$ so we compare the given series with the series $\sum 5/(2n^2)$. Observe that

$$\frac{5}{2n^2 + 4n + 3} < \frac{5}{2n^2}$$

because the left side has a bigger denominator. (In the notation of the Comparison Test, a_n is the left side and b_n is the right side.) We know that

$$\sum_{n=1}^{\infty} \frac{5}{2n^2} = \frac{5}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

is convergent because it's a constant times a p -series with $p = 2 > 1$. Therefore

$$\sum_{n=1}^{\infty} \frac{5}{2n^2 + 4n + 3}$$

is convergent by part (i) of the Comparison Test.

EXAMPLE 2 Test the series $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ for convergence or divergence.

SOLUTION This series was tested (using the Integral Test) but it is also possible to test it by comparing it with the harmonic series. Observe that $\ln n > 1$ for $n \geq 3$ and so

$$\frac{\ln n}{n} > \frac{1}{n} \quad n \geq 3$$

We know that $\sum 1/n$ is divergent (p -series with $p = 1$). Thus, the given series is divergent by the Comparison Test.

NOTE 2 • The terms of the series being tested must be smaller than those of a convergent series or larger than those of a divergent series. If the terms are larger than the terms of a convergent series or smaller than those of a divergent series, then the Comparison Test doesn't apply. Consider, for instance, the series

$$\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$$

The inequality

$$\frac{1}{2^n - 1} > \frac{1}{2^n}$$

is useless as far as the Comparison Test is concerned because $\sum b_n = \sum \left(\frac{1}{2}\right)^n$ is convergent and $a_n > b_n$. Nonetheless, we have the feeling that $\sum 1/(2^n - 1)$ ought to be convergent because it is very similar to the convergent geometric series $\sum \left(\frac{1}{2}\right)^n$. In such cases the following test can be used.

The Limit Comparison Test Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

EXAMPLE 3 Test the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ for convergence or divergence.

SOLUTION We use the Limit Comparison Test with

$$a_n = \frac{1}{2^n - 1} \qquad b_n = \frac{1}{2^n}$$

and obtain

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1/(2^n - 1)}{1/2^n} = \lim_{n \rightarrow \infty} \frac{2^n}{2^n - 1} = \lim_{n \rightarrow \infty} \frac{1}{1 - 1/2^n} = 1 > 0$$

Since this limit exists and $\sum 1/2^n$ is a convergent geometric series, the given series converges by the Limit Comparison Test.

EXAMPLE 4 Determine whether the series $\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{5 + n^5}}$ converges or diverges.

SOLUTION The dominant part of the numerator is $2n^2$ and the dominant part of the denominator is $\sqrt{n^5} = n^{5/2}$. This suggests taking

$$a_n = \frac{2n^2 + 3n}{\sqrt{5 + n^5}} \qquad b_n = \frac{2n^2}{n^{5/2}} = \frac{2}{n^{1/2}}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{2n^2 + 3n}{\sqrt{5 + n^5}} \cdot \frac{n^{1/2}}{2} = \lim_{n \rightarrow \infty} \frac{2n^{5/2} + 3n^{3/2}}{2\sqrt{5 + n^5}} \\ &= \lim_{n \rightarrow \infty} \frac{2 + \frac{3}{n}}{2\sqrt{\frac{5}{n^5} + 1}} = \frac{2 + 0}{2\sqrt{0 + 1}} = 1 \end{aligned}$$

Since $\sum b_n = 2 \sum 1/n^{1/2}$ is divergent (p -series with $p = \frac{1}{2} < 1$), the given series diverges by the Limit Comparison Test.

Alternating Series

The convergence tests that we have looked at so far apply only to series with positive terms. In this section and the next we learn how to deal with series whose terms are not necessarily positive. Of particular importance are *alternating series*, whose terms alternate in sign.

An **alternating series** is a series whose terms are alternately positive and negative. Here are two examples:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$
$$-\frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \frac{6}{7} - \cdots = \sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1}$$

We see from these examples that the n th term of an alternating series is of the form

$$a_n = (-1)^{n-1}b_n \quad \text{or} \quad a_n = (-1)^nb_n$$

where b_n is a positive number. (In fact, $b_n = |a_n|$.)

The following test says that if the terms of an alternating series decrease toward 0 in absolute value, then the series converges.

The Alternating Series Test If the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \cdots \quad (b_n > 0)$$

satisfies

- (i) $b_{n+1} \leq b_n$ for all n
- (ii) $\lim_{n \rightarrow \infty} b_n = 0$

then the series is convergent.

EXAMPLE 1 The alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$

satisfies

$$(i) \quad b_{n+1} < b_n \quad \text{because} \quad \frac{1}{n+1} < \frac{1}{n}$$

$$(ii) \quad \lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

so the series is convergent by the Alternating Series Test.

EXAMPLE 2 The series $\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n-1}$ is alternating but

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{3n}{4n-1} = \lim_{n \rightarrow \infty} \frac{3}{4 - \frac{1}{n}} = \frac{3}{4}$$

so condition (ii) is not satisfied. Instead, we look at the limit of the n th term of the series:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{(-1)^n 3n}{4n-1}$$

This limit does not exist, so the series diverges by the Test for Divergence.

EXAMPLE 3 Test the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3 + 1}$ for convergence or divergence.

SOLUTION The given series is alternating so we try to verify conditions (i) and (ii) of the Alternating Series Test.

Unlike the situation in Example 1, it is not obvious that the sequence given by $b_n = n^2/(n^3 + 1)$ is decreasing. However, if we consider the related function

$f(x) = x^2/(x^3 + 1)$, we find that

$$f'(x) = \frac{x(2 - x^3)}{(x^3 + 1)^2}$$

Since we are considering only positive x , we see that $f'(x) < 0$ if $2 - x^3 < 0$, that is, $x > \sqrt[3]{2}$. Thus, f is decreasing on the interval $(\sqrt[3]{2}, \infty)$. This means that $f(n + 1) < f(n)$ and therefore $b_{n+1} < b_n$ when $n \geq 2$. (The inequality $b_2 < b_1$ can be verified directly but all that really matters is that the sequence $\{b_n\}$ is eventually decreasing.)

Condition (ii) is readily verified:

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{n^2}{n^3 + 1} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1 + \frac{1}{n^3}} = 0$$

Thus, the given series is convergent by the Alternating Series Test.

Absolute Convergence and the Ratio and Root Tests

Given any series $\sum a_n$, we can consider the corresponding series

$$\sum_{n=1}^{\infty} |a_n| = |a_1| + |a_2| + |a_3| + \cdots$$

whose terms are the absolute values of the terms of the original series.

I Definition A series $\sum a_n$ is called absolutely convergent if the series of absolute values $\sum |a_n|$ is convergent.

Notice that if $\sum a_n$ is a series with positive terms, then $|a_n| = a_n$ and so absolute convergence is the same as convergence in this case.

EXAMPLE 1 The series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$$

is absolutely convergent because

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n^2} \right| = \sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$$

is a convergent p -series ($p = 2$).

EXAMPLE 2 We know that the alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

is convergent, but it is not absolutely convergent because the corresponding series of absolute values is

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

which is the harmonic series (p -series with $p = 1$) and is therefore divergent.

2 Definition A series $\sum a_n$ is called conditionally convergent if it is convergent but not absolutely convergent.

Example 2 shows that the alternating harmonic series is conditionally convergent. Thus, it is possible for a series to be convergent but not absolutely convergent. However, the next theorem shows that absolute convergence implies convergence.

3 Theorem If a series $\sum a_n$ is absolutely convergent, then it is convergent.

EXAMPLE 3 Determine whether the series

$$\sum_{n=1}^{\infty} \frac{\cos n}{n^2} = \frac{\cos 1}{1^2} + \frac{\cos 2}{2^2} + \frac{\cos 3}{3^2} + \cdots$$

is convergent or divergent.

SOLUTION This series has both positive and negative terms, but it is not alternating.

(The signs change irregularly.) We can apply the Comparison Test to the series of absolute values

$$\sum_{n=1}^{\infty} \left| \frac{\cos n}{n^2} \right| = \sum_{n=1}^{\infty} \frac{|\cos n|}{n^2}$$

Since $|\cos n| \leq 1$ for all n , we have

$$\frac{|\cos n|}{n^2} \leq \frac{1}{n^2}$$

We know that $\sum 1/n^2$ is convergent (p -series with $p = 2$) and therefore $\sum |\cos n|/n^2$ is convergent by the Comparison Test. Thus, the given series $\sum (\cos n)/n^2$ is absolutely convergent and therefore convergent by Theorem 3.

The following test is very useful in determining whether a given series is absolutely convergent.

The Ratio Test

(i) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent
(and therefore convergent).

(ii) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L > 1$ or $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$, then the series $\sum_{n=1}^{\infty} a_n$
is divergent.

(iii) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, the Ratio Test is inconclusive; that is, no conclusion can
be drawn about the convergence or divergence of $\sum a_n$.

EXAMPLE 4 Test the series $\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{3^n}$ for absolute convergence.

SOLUTION We use the Ratio Test with $a_n = (-1)^n n^3/3^n$:

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{\frac{(-1)^{n+1}(n+1)^3}{3^{n+1}}}{\frac{(-1)^n n^3}{3^n}} \right| = \frac{(n+1)^3}{3^{n+1}} \cdot \frac{3^n}{n^3} \\ &= \frac{1}{3} \left(\frac{n+1}{n} \right)^3 = \frac{1}{3} \left(1 + \frac{1}{n} \right)^3 \rightarrow \frac{1}{3} < 1 \end{aligned}$$

Thus, by the Ratio Test, the given series is absolutely convergent and therefore convergent.

EXAMPLE 5 Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n^n}{n!}$.

SOLUTION Since the terms $a_n = n^n/n!$ are positive, we don't need the absolute value signs.

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n} = \frac{(n+1)(n+1)^n}{(n+1)n!} \cdot \frac{n!}{n^n} \\ &= \left(\frac{n+1}{n}\right)^n = \left(1 + \frac{1}{n}\right)^n \rightarrow e \quad \text{as } n \rightarrow \infty\end{aligned}$$

Since $e > 1$, the given series is divergent by the Ratio Test.

NOTE ◦ Although the Ratio Test works in Example 5, an easier method is to use the Test for Divergence. Since

$$a_n = \frac{n^n}{n!} = \frac{n \cdot n \cdot n \cdot \cdots \cdot n}{1 \cdot 2 \cdot 3 \cdot \cdots \cdot n} \geq n$$

it follows that a_n does not approach 0 as $n \rightarrow \infty$. Therefore, the given series is divergent by the Test for Divergence.

The following test is convenient to apply when n th powers occur.

The Root Test

(i) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (and therefore convergent).

(ii) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L > 1$ or $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

(iii) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$, the Root Test is inconclusive.

If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$, then part (iii) of the Root Test says that the test gives no information. The series $\sum a_n$ could converge or diverge. (If $L = 1$ in the Ratio Test, don't try the Root Test because L will again be 1.)

EXAMPLE 6 Test the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$.

SOLUTION

$$a_n = \left(\frac{2n+3}{3n+2} \right)^n$$

$$\sqrt[n]{|a_n|} = \frac{2n+3}{3n+2} = \frac{2 + \frac{3}{n}}{3 + \frac{2}{n}} \rightarrow \frac{2}{3} < 1$$

Thus, the given series converges by the Root Test.

EXAMPLE

Which of the following series converge? Give reasons for your answers.

(a) $\sum_{n=1}^{\infty} \frac{1}{2^n + 1}$, (b) $\sum_{n=1}^{\infty} \frac{3n + 1}{n^3 + 1}$, (c) $\sum_{n=2}^{\infty} \frac{1}{\ln n}$.

Solution In each case we must find a suitable comparison series that we already know converges or diverges.

(a) Since $0 < \frac{1}{2^n + 1} < \frac{1}{2^n}$ for $n = 1, 2, 3, \dots$, and since $\sum_{n=1}^{\infty} \frac{1}{2^n}$ is a convergent geometric series, the series $\sum_{n=1}^{\infty} \frac{1}{2^n + 1}$ also converges by comparison.

(b) Observe that $\frac{3n + 1}{n^3 + 1}$ behaves like $\frac{3}{n^2}$ for large n , so we would expect to compare the series with the convergent p -series $\sum_{n=1}^{\infty} n^{-2}$. We have, for $n \geq 1$,

$$\frac{3n + 1}{n^3 + 1} = \frac{3n}{n^3 + 1} + \frac{1}{n^3 + 1} < \frac{3n}{n^3} + \frac{1}{n^3} < \frac{3}{n^2} + \frac{1}{n^2} = \frac{4}{n^2}.$$

Thus, the given series converges

(c) For $n = 2, 3, 4, \dots$, we have $0 < \ln n < n$. Thus $\frac{1}{\ln n} > \frac{1}{n}$. Since $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges to infinity (it is a harmonic series), so does $\sum_{n=2}^{\infty} \frac{1}{\ln n}$ by comparison.

EXAMPLE

Which of the following series converge? Give reasons for your answers.

$$(a) \sum_{n=1}^{\infty} \frac{1}{1 + \sqrt{n}}, \quad (b) \sum_{n=1}^{\infty} \frac{n + 5}{n^3 - 2n + 3}.$$

Solution Again we must make appropriate choices for comparison series.

(a) The terms of this series decrease like $1/\sqrt{n}$. Observe that

$$L = \lim_{n \rightarrow \infty} \frac{\frac{1}{1 + \sqrt{n}}}{\frac{1}{\sqrt{n}}} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{1 + \sqrt{n}} = \lim_{n \rightarrow \infty} \frac{1}{(1/\sqrt{n}) + 1} = 1.$$

Since the p -series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ diverges to infinity ($p = 1/2$), so does the series

$\sum_{n=1}^{\infty} \frac{1}{1 + \sqrt{n}}$, by the limit comparison test.

(b) For large n , the terms behave like n/n^3 , so let us compare the series with the p -series $\sum_{n=1}^{\infty} 1/n^2$, which we know converges.

$$L = \lim_{n \rightarrow \infty} \frac{\frac{n + 5}{n^3 - 2n + 3}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{n^3 + 5n^2}{n^3 - 2n + 3} = 1.$$

Since $L < \infty$, the series $\sum_{n=1}^{\infty} \frac{n + 5}{n^3 - 2n + 3}$ also converges by the limit comparison test.

EXAMPLE

Test the series $\sum_{n=1}^{\infty} \frac{1 + \sin n}{n^2}$ for convergence.

Solution Since

$$\lim_{n \rightarrow \infty} \frac{\frac{1 + \sin n}{n^2}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} (1 + \sin n)$$

does not exist, the limit comparison test gives us no information. However, since $\sin n \leq 1$, we have

$$0 \leq \frac{1 + \sin n}{n^2} \leq \frac{2}{n^2} \quad \text{for } n = 1, 2, 3, \dots$$

The given series does, in fact, converge by comparison with $\sum_{n=1}^{\infty} 1/n^2$, using the ordinary comparison test.

EXAMPLE

Test the following series for convergence:

$$(a) \sum_{n=1}^{\infty} \frac{99^n}{n!}, \quad (b) \sum_{n=1}^{\infty} \frac{n^5}{2^n}, \quad (c) \sum_{n=1}^{\infty} \frac{n!}{n^n}, \quad (d) \sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2}.$$

Solution We use the ratio test for each of these series.

$$(a) \rho = \lim_{n \rightarrow \infty} \frac{99^{n+1}}{(n+1)!} \bigg/ \frac{99^n}{n!} = \lim_{n \rightarrow \infty} \frac{99}{n+1} = 0 < 1.$$

Thus $\sum_{n=1}^{\infty} (99^n/n!)$ converges.

$$(b) \rho = \lim_{n \rightarrow \infty} \frac{(n+1)^5}{2^{n+1}} \bigg/ \frac{n^5}{2^n} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{n+1}{n} \right)^5 = \frac{1}{2} < 1.$$

Hence $\sum_{n=1}^{\infty} (n^5/2^n)$ converges.

$$(c) \rho = \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \bigg/ \frac{n!}{n^n} = \lim_{n \rightarrow \infty} \frac{(n+1)!n^n}{(n+1)^{n+1}n!} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n \\ = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n} \right)^n} = \frac{1}{e} < 1.$$

Thus $\sum_{n=1}^{\infty} (n!/n^n)$ converges.

$$(d) \rho = \lim_{n \rightarrow \infty} \frac{(2(n+1))!}{((n+1)!)^2} \bigg/ \frac{(2n)!}{(n!)^2} = \lim_{n \rightarrow \infty} \frac{(2n+2)(2n+1)}{(n+1)^2} = 4 > 1.$$

Thus $\sum_{n=1}^{\infty} (2n)!/(n!)^2$ diverges to infinity.

Ex: Consider the series with terms

$$a_n = \begin{cases} n/2^n & , n \text{ odd} \\ 1/2^n & , n \text{ even} \end{cases} .$$

Does $\sum a_n$ converge?

we apply the Root Test, finding that

$$\sqrt[n]{a_n} = \begin{cases} \sqrt[n]{n}/2 & , n \text{ odd} \\ 1/2 & , n \text{ even} \end{cases}$$

Thus,

$$\frac{1}{2} \leq \sqrt[n]{a_n} \leq \frac{\sqrt[n]{n}}{2} .$$

Since $\sqrt[n]{n} \rightarrow 1$, we have $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = 1/2$ (by Sandwich theorem)

The limit is less than 1, so the series converges by Root Test.

Ex: Which of the following series converge and which diverge?

a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

b) $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$

c) $\sum_{n=1}^{\infty} \left(\frac{1}{1+n} \right)^n$

we apply the Root Test to each series

a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ converges because $\sqrt[n]{\frac{n^2}{2^n}} = \frac{\sqrt[n]{n^2}}{\sqrt[n]{2^n}} = \frac{(\sqrt[n]{n})^2}{2} \rightarrow \frac{1^2}{2} < 1$

b) $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$ diverges because $\sqrt[n]{\frac{2^n}{n^3}} = \frac{2}{\sqrt[n]{n^3}} = \frac{2}{(\sqrt[n]{n})^3} \rightarrow \frac{2}{1^3} > 1$

c) $\sum_{n=1}^{\infty} \left(\frac{1}{1+n} \right)^n$ converges because $\sqrt[n]{\left(\frac{1}{1+n} \right)^n} = \frac{1}{1+n} \rightarrow 0 < 1$

$$\textcircled{1} \sum_{n=1}^{\infty} \frac{n! (2n)!}{(3n)!}$$

$$a_n = \frac{n! (2n)!}{(3n)!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{\frac{(n+1)! (2(n+1))!}{3(n+1)!}}{\frac{n! (2n)!}{(3n)!}} = \lim_{n \rightarrow \infty} \frac{(n+1)! (2n+2)! (3n)!}{(3n+3)! n! (2n)!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\cancel{n!} (n+1) \cancel{(2n)!} (2n+1)(2n+2) \cancel{(3n)!}}{(3n)! \cancel{(3n+1)} (3n+2)(3n+3) \cancel{n!} \cancel{(2n)!}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)(2n+1)(2n+2)}{(3n+1)(3n+2)(3n+3)} \right| = \lim_{n \rightarrow \infty} \frac{\cancel{n} (1+\frac{1}{n}) \cancel{n} (2+\frac{1}{n}) \cancel{n} (2+\frac{2}{n})}{\cancel{n} (3+\frac{1}{n}) \cancel{n} (3+\frac{2}{n}) \cancel{n} (3+\frac{3}{n})}$$

$$\lim_{n \rightarrow \infty} \frac{1 \cdot 2 \cdot 2}{3 \cdot 3 \cdot 3} = \frac{4}{27} < 1 \Rightarrow \text{convergent}$$

$$(2) \sum_{n=1}^{\infty} \frac{3^n}{5^n(n+2)}$$

$$a_n = \frac{3^n}{5^n(n+2)}$$

$$L = \lim_{n \rightarrow \infty} (a_n)^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{3^n}{5^n(n+2)} \right)^{1/n} = \lim_{n \rightarrow \infty} \frac{3^{n/n}}{5^{n/n}(n+2)^{1/n}}$$

$$= \frac{3}{5} \lim_{n \rightarrow \infty} \frac{1}{(n+2)^{1/n}} \quad (\infty^0)$$

$$(n+2)^{1/n} = e^{1/n \ln(n+2)} \Rightarrow \lim_{n \rightarrow \infty} \frac{\ln(n+2)}{n} = \frac{\infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\frac{n+2}{1}} = \frac{1}{n+2} \rightarrow 0$$

convergent

$$e^0 = 1 \quad L = \frac{3}{5} < 1 \Rightarrow$$

EXAMPLE

Test the following series for absolute convergence:

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1}, \quad (b) \sum_{n=1}^{\infty} \frac{n \cos(n\pi)}{2^n}.$$

Solution

$$(a) \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n-1}}{2n-1} \right| \bigg/ \frac{1}{n} = \lim_{n \rightarrow \infty} \frac{n}{2n-1} = \frac{1}{2} > 0.$$

Since the harmonic series $\sum_{n=1}^{\infty} (1/n)$ diverges to infinity, the comparison test assures us that $\sum_{n=1}^{\infty} ((-1)^{n-1}/(2n-1))$ does not converge absolutely.

$$(b) \rho = \lim_{n \rightarrow \infty} \left| \frac{(n+1) \cos((n+1)\pi)}{2^{n+1}} \bigg/ \frac{n \cos(n\pi)}{2^n} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2} < 1.$$

(Note that $\cos(n\pi)$ is just a fancy way of writing $(-1)^n$.) Therefore (ratio test) $\sum_{n=1}^{\infty} ((n \cos(n\pi))/2^n)$ converges absolutely.

EXAMPLE

Test the following series for absolute and conditional convergence:

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}, \quad (b) \sum_{n=2}^{\infty} \frac{\cos(n\pi)}{\ln n}, \quad (c) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^4}.$$

Solution The absolute values of the terms in series (a) and (b) are $1/n$ and $1/(\ln n)$, respectively. Since $1/(\ln n) > 1/n$, and $\sum_{n=1}^{\infty} 1/n$ diverges to infinity, neither series (a) nor (b) converges absolutely. However, both series satisfy the requirements of Theorem 11.2 and so both converge. Each of these series is conditionally convergent.

Alter.series test
Series (c) is absolutely convergent because $|(-1)^{n-1}/n^4| = 1/n^4$, and $\sum_{n=1}^{\infty} 1/n^4$ is a convergent p -series ($p = 4 > 1$). We could establish its convergence using Theorem 11.2, but there is no need to do that since every absolutely convergent series is Alter.series test convergent.

EXAMPLE

Test the following series for convergence:

(a) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n+1}{n},$

(b) $1 - \frac{1}{4} + \frac{1}{3} - \frac{1}{16} + \frac{1}{5} - \cdots = \sum_{n=1}^{\infty} a_n,$ where

$$a_n = \begin{cases} 1/n & \text{if } n \text{ is odd,} \\ -1/n^2 & \text{if } n \text{ is even.} \end{cases}$$

Solution

- (a) Here, the terms a_n alternate and decrease in size as n increases. However, $\lim_{n \rightarrow \infty} |a_n| = 1 \neq 0$. The alternating series test does not apply. In fact, the given series diverges because its terms do not approach 0.
- (b) This series alternates and its terms have limit zero. However, the terms are not decreasing in size (even ultimately). Once again, the alternating series test cannot be applied. In fact, since

$$\begin{aligned} -\frac{1}{4} - \frac{1}{16} - \cdots - \frac{1}{(2n)^2} - \cdots & \quad \text{converges, and} \\ 1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1} + \cdots & \quad \text{diverges to infinity,} \end{aligned}$$

it is readily seen that the given series diverges to infinity.